Market Research

Market Research Report: U.S. Garden App Market (2026)

Prepared for: Product Management Team

Prepared by: Senior Market Research Analyst

Date: February 22, 2026

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1. Executive Summary

The U.S. gardening market is undergoing a generational shift. With **18.3 million new gardeners** entering the market since 2020 and **Millennials/Gen Z** now representing the fastest-growing demographics, the demand for "smart" guidance is at an all-time high. While the broader market is valued at **$22B–$136B**, the digital "helper" niche is underserved in high-accuracy, hyper-local guidance.

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2. Task 1: Market Segmentation (U.S. Region)

Market Segment	Brief Description	Estimated Market Size	Total Addressable Market (TAM)	Key Needs
Urban "Small Space" Gardeners	Millennials/Gen Z in apartments/condos using balconies, windowsills, or rooftops.	**$3.5B** (Smart Indoor Market, 2025). CAGR: **13.5%**.	**$1.05B** (Methodology A)	Space optimization and "Succession Planting" to maximize tiny yields.

Suburban "Newbie" Homeowners	First-time homeowners (ages 25-40) managing their first 1/4 acre or raised beds.	**$21.8B** (U.S. Residential Tools/Seeds, 2024). CAGR: **6.0%**.	**$1.8B** (Methodology B)	Overcoming "Brown Thumb" anxiety and automated maintenance schedules.
"Grow Your Own" Enthusiasts	Consumers focused on edible crops (vegetables/herbs) for health and sustainability.	**$1.2B** (Smart Sensors/Tech, 2023). CAGR: **12%**.	**$630M** (Methodology C)	Pest/disease identification and harvesting timelines for food safety.

TAM Calculation Methodology

Calculated using a **Bottom-Up Approach** based on current 2026 penetration data:

- **Total US Households:** ~131,000,000
- **Active Gardening Households (55%):** ~72,000,000
- **Annual Subscription Target:** $40.00
- **Formula:** (72M households) x ($40.00 annual sub) = **$2.88 Billion**

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3. Task 2: Competitive Analysis

Competitive Landscape

Competitor	Strengths	Weaknesses	Est. Market Share	Key Features
PlantIn / PictureThis	Best-in-class AI Image ID.	High subscription friction; generic care advice.	~25-30%	Disease Diagnosis, Botanist Chat.
Planta	Highly polished UI; indoor focus.	Poor outdoor/hardiness zone integration.	~15%	Smart Watering Reminders.

Seed to Spoon	Deep library for edibles; organic focus.	Steeper learning curve; UI feels dated.	~5%	Companion Planting Guides.

Identified Gaps & Unmet Needs

1. **Hyper-Local Logic:** Users express frustration with generic "Hardiness Zone" advice. They need micro-climate data (e.g., "It's raining at your house today, skip watering").
2. **Succession Planning:** Beginners lack knowledge on what to plant next after a harvest.
3. **The "Why" Factor:** Current apps lack empathetic coaching to help users understand *why* a plant is struggling before it dies.

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4. Strategic Evaluation

- **Unique Value Proposition (UVP):** "The Empathetic Gardener"—predictive maintenance that adapts to real-time local weather and succession coaching.
- **Scalability:** The **Urban/Small-space** segment is the strongest for growth (13.5% CAGR).
- **User Pain Points:** The most acute pain point is **"The Fear of Killing."** Social media sentiment (Reddit r/gardening) indicates a need for a "single source of truth" to manage mental load.

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5. Final Recommendation

Initial Target Segment: The "Suburban Newbie" Homeowner.

Rationale: This segment has the highest **Total Addressable Market ($1.8B+)** and the highest average annual spend on gardening ($616/year). These users have high "sunk costs" in their yards and are more likely to pay for a subscription to protect their investment compared to casual urban growers.

User Personas & JTBD

User Mapping & JTBD

Task 1: User Personas

1. Maya: The "Suburban Sanctuary Seeker"

- **Photo Placeholder:** *[A 30-something professional standing in a sun-drenched, slightly messy backyard with several potted hydrangeas and a raised bed.]*
- **Role & Responsibilities:** Marketing Manager; accountable for a high-stress 40-hour work week and the recent "brand management" of her first home's curb appeal.
- **Demographics:** 32 years old, recently moved from a city apartment to a suburb of Philadelphia. Tech-literate, relies on apps for fitness and meal planning. Zero previous gardening experience.
- **Goals & Motivations:** Wants a "Pinterest-ready" outdoor space for hosting friends; seeks a therapeutic hobby to decompress from work; wants to ensure the $1,200 she spent at the local nursery doesn't go to waste.
- **Frustrations & Pain Points:** Overwhelmed by "Hardiness Zones"—she doesn't understand why a plant at the store says "Zone 7" but dies in her yard. She forgets to water during busy work weeks but over-corrects on weekends, leading to root rot.
- **A Quote:** *"I spent my entire Saturday and half my paycheck at the nursery. I just need an app that tells me exactly what to do so I don't kill everything by July."*

2. Liam: The "Health-Conscious Harvester"

- **Photo Placeholder:** *[A late-30s father in a t-shirt and gardening gloves, holding a toddler who is reaching for a cherry tomato.]*
- **Role & Responsibilities:** High School Teacher; accountable for family nutrition, household budget, and "weekend warrior" DIY projects.
- **Demographics:** 38 years old, living in Austin, TX. Values sustainability and organic living. Has attempted a small veggie patch before with "mixed results" (mostly weeds and one very sad cucumber).
- **Goals & Motivations:** Wants to teach his children where food comes from; wants to reduce his grocery bill by growing high-yield herbs and greens; motivated by the "gamification" of seeing a plant grow from seed to table.
- **Frustrations & Pain Points:** Pest identification—he sees a bug and doesn't know if it's a "good" ladybug or a "bad" aphid. He struggles with "Succession Planting"; once his spring lettuce bolts in the Texas heat, his garden sits empty for the rest of the year.
- **A Quote:** *"I want my kids to eat a tomato they actually grew, but between the heatwaves and the bugs, I feel like I'm playing a guessing game I'm destined to lose."*

Task 2: Jobs-to-be-Done (JTBD)

Priority	Job to be Done	Acceptance Criteria

P0	**As a Suburban Newbie (Maya),** I want to scan my existing plants and receive a personalized maintenance calendar **so that** I can stop worrying about "when" and "how much" to water or prune.	• App identifies plant species with >95% accuracy. • Generates a dynamic calendar that syncs with phone notifications. • Watering alerts adjust automatically based on local rain forecasts.
P0	**As a First-time Gardener (Maya/Liam),** I want to receive "early warning" alerts for common pests or diseases based on my specific micro-climate **so that** I can intervene before my plants die.	• Push notifications sent when local humidity/temp spikes (fungus risk). • Photo-upload feature identifies pests and suggests organic remedies. • Provides a "Health Score" for each plant in the garden.

P1	**As a Sustainable Harvester (Liam),** I want to be told exactly what to plant "next" based on my current harvest and local season **so that** my garden stays productive year-round.	• Suggests "Succession Plants" (e.g., plant kale after peas). • Calculates "Days to Harvest" based on the user's specific zip-code weather. • Links to buy seeds/starts that are appropriate for the current date.
P2	**As an Unsure Newbie (Maya),** I want to understand the "Why" behind a care instruction (e.g., *why* water at the base?) **so that** I can actually learn the skill of gardening rather than just following orders.	• "Care Tips" include 1-sentence botanical explanations. • Tooltips explain terms like "Deadheading" or "Bolting" in plain English. • Short (<30 sec) video snippets demonstrate tricky techniques.

Strategy Recommendation

To win this market, we must move beyond the "Plant Encyclopedia" model. **P0 tasks must focus on "Cognitive Offloading."** Maya and Liam don't want to become botanists overnight; they want a co-pilot that manages the mental load of the garden.

Our MVP should prioritize the **Hyper-Local Weather Integration** and the **Maintenance Calendar**, as these directly address the "Fear of Killing" identified in our research.

Next Step: I recommend we move into low-fidelity wireframing for the "Maintenance Dashboard" to see how Maya would interact with her daily tasks. Shall I begin the wireframe requirements?

PRD

PRD

Product Requirements Document: "The Empathetic Gardener" V1.0

Status: Draft / In-Progress

Owner: Senior PM / PRD Assistant

Last Updated: February 22, 2026

Tab 1: Project Overview

1.1 Problem Statement

First-time suburban homeowners ("Newbies") are entering the gardening market at record rates but face high "failure anxiety." Current solutions provide generic botanical data rather than empathetic, hyper-local guidance. This leads to dead plants, wasted money (avg. $616/year), and abandoned hobbies.

1.2 Product Goals

- **Goal 1:** Reduce the "Fear of Killing" by providing a reliable, automated maintenance co-pilot.
- **Goal 2:** Build a "Single Source of Truth" for home garden management.
- **Goal 3:** Establish a sustainable revenue stream through a tiered freemium model.

1.3 Success Metrics (KPIs)

- **Retention (North Star):** % of users who remain active 30 days after their first "Frost Date" or "Heat Wave" alert.
- **Engagement:** Avg. number of "Maintenance Tasks" marked complete per user per week.
- **Conversion:** % of free users who upgrade to Premium for "Expert Review" or "Advanced AI."
- **Monetization:** Ad impressions per Daily Active User (DAU).

Tab 2: User Research & Personas

2.1 Target Personas

- **Maya (The Suburban Sanctuary Seeker):** High sunk cost in landscaping; needs automated peace of mind.
- **Liam (The Health-Conscious Harvester):** Focus on edibles; needs pest ID and yield optimization.

2.2 Jobs-to-be-Done (JTBD)

- **P0:** "As a newbie, I want to scan my plant so I know exactly what it is and how to keep it alive today."
- **P0:** "As a busy professional, I want a calendar that tells me when to water based on the *actual* rain in my yard."

Tab 3: Functional Requirements

ID	Feature	Priority	Description	Acceptance Criteria
FR1	**Third-Party Plant ID**	**P0**	Integration with a 3rd party API (e.g., Plant.id) for instant species and health identification.	ID accuracy > 90%; Results delivered < 3 seconds.
FR2	**Weather-Sync Calendar**	**P0**	A maintenance schedule that adjusts based on software-based weather API data.	Calendar updates triggers (Water/Cover/Harvest) based on real-time precipitation/temp data.
FR3	**Ad-Supported Free Tier**	**P1**	Implementation of interstitial or banner ads for non-paying users.	Ads do not interrupt the "Scanner" camera view.
FR4	**Premium: Expert Review**	**P1**	A paywalled feature allowing users to send photos to human botanists for complex issues.	Response time < 24 hours; Secure payment gateway integration.
FR5	**Advanced AI Insights**	**P2**	Paywalled predictive insights (e.g., "Based on	[[TBD]] - Need to define specific "Advanced" parameters.

			growth rate, your tomato will be ready in 12 days").	
FR6	**Companion Planting**	**P2**	Logic to suggest what to plant next to existing inventory.	Link plants in database via "Compatibility" tags.

Tab 4: Technical & Business Logic

4.1 Tech Stack (V1)

- **Plant Identification:** External API (Plant.id or PlantNet).
- **Weather Data:** OpenWeatherMap API (Software-only, no hardware integration).
- **Frontend:** Cross-platform (Flutter or React Native).
- **Database:** [[TBD - Cloud-based user garden profiles]].

4.2 Monetization Model (Freemium)

- **Free Tier:** Unlimited basic ID scans, standard weather alerts, ad-supported experience.
- **Premium Tier:** Ad-free, Expert Human Review credits, and "Advanced AI" (Succession planting & yield prediction).

Tab 5: Roadmap & Gaps

5.1 Open Questions [[TBD]]

- **Ad Provider:** Which ad network (AdMob, etc.) provides the best CPM for the gardening/home-improvement niche?
- **Expert Sourcing:** How will we recruit the "Human Experts" for the Premium tier? Will they be internal staff or a gig-economy network?
- **Advanced AI Definition:** We need to define exactly which AI features are "Advanced" enough to justify the paywall vs. the basic scanner.

5.2 Next Steps

1. Finalize the choice of **Weather API** (comparing OpenWeather vs. AccuWeather for accuracy).
2. Define the **Ad-frequency strategy** (to ensure Maya doesn't get frustrated while trying to save a dying plant!).
3. Draft **UI/UX Wireframes** for the "Maintenance Dashboard."

Roadmap

Roadmap html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Garden App MVP Roadmap</title>
   <script src="https://cdn.tailwindcss.com"></script>
   <link rel="preconnect" href="https://fonts.googleapis.com">
   <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
rel="stylesheet">
   <style>
      body { font-family: 'Inter', sans-serif; background-color: #f8fafc; }
      .roadmap-container { max-width: 1200px; margin: 2rem auto; padding: 0 1rem; }
      .phase-card { background: white; border-radius: 1rem; box-shadow: 0 4px 6px -1px
rgba(0,0,0,0.1); border-top: 5px solid; transition: transform 0.2s; }
      .phase-card:hover { transform: translateY(-4px); }
      .dev-border { border-color: #4f46e5; }
      .pre-border { border-color: #f59e0b; }
      .post-border { border-color: #10b981; }
      .milestone-item { position: relative; padding-left: 1.5rem; margin-bottom: 0.75rem; }
      .milestone-item::before { content: ''; position: absolute; left: 0; top: 0.5rem; width: 8px;
height: 8px; border-radius: 50%; background: #cbd5e1; }
      .priority-p0 { color: #dc2626; font-weight: 700; font-size: 0.7rem; }
      @media print { #header { display: none; } .roadmap-grid { grid-template-columns: 1fr
!important; } }
   </style>
</head>
<body class="p-4">
   <header id="header" class="flex justify-between items-center mb-10 roadmap-container">
      <div>
         <h1 class="text-3xl font-bold text-slate-800 tracking-tight">MVP Product Roadmap</h1>
         <p class="text-slate-500">Project: The Empathetic Gardener v1.0</p>
      </div>
      <button onclick="window.print()" class="bg-slate-800 text-white px-5 py-2 rounded-lg
font-semibold hover:bg-slate-700 transition">Download PDF</button>
   </header>

   <main class="roadmap-container">
      <div class="grid grid-cols-1 lg:grid-cols-3 gap-8 roadmap-grid">
```

```
<section class="phase-card dev-border p-6">
  <div class="flex items-center justify-between mb-4">
    <h2 class="text-xl font-bold text-slate-800">1. Development</h2>
    <span class="bg-indigo-100 text-indigo-700 text-xs px-2 py-1 rounded">Months
1-3</span>
  </div>
  <p class="text-sm text-slate-500 mb-6">Focus on core functionality and API
integration.</p>

  <div class="space-y-4">
    <div class="milestone-item">
      <span class="priority-p0">P0</span>
      <p class="text-sm font-semibold">API Pipeline Setup</p>
      <p class="text-xs text-slate-400">Integrate Plant.id & OpenWeather APIs.</p>
    </div>
    <div class="milestone-item">
      <span class="priority-p0">P0</span>
      <p class="text-sm font-semibold">Maintenance Logic Engine</p>
      <p class="text-xs text-slate-400">Rule-based watering/pruning triggers.</p>
    </div>
    <div class="milestone-item">
      <span class="text-amber-600 font-bold text-[0.7rem]">P1</span>
      <p class="text-sm font-semibold">Ad-SDK Implementation</p>
      <p class="text-xs text-slate-400">Integration with Google AdMob for
Freemium.</p>
    </div>
    <div class="milestone-item">
      <span class="text-amber-600 font-bold text-[0.7rem]">P1</span>
      <p class="text-sm font-semibold">User Profile & Garden DB</p>
      <p class="text-xs text-slate-400">Cloud storage for Maya's plant inventory.</p>
    </div>
  </div>
</section>

<section class="phase-card pre-border p-6">
  <div class="flex items-center justify-between mb-4">
    <h2 class="text-xl font-bold text-slate-800">2. Pre-Release</h2>
    <span class="bg-amber-100 text-amber-700 text-xs px-2 py-1 rounded">Month
4</span>
  </div>
  <p class="text-sm text-slate-500 mb-6">QA, Beta testing, and marketing
preparation.</p>

  <div class="space-y-4">
```

```
<div class="milestone-item">
    <span class="priority-p0">P0</span>
    <p class="text-sm
```

Business Process Flow

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Garden App Process Flow Strategy</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap
" rel="stylesheet">
    <style>
        /* Base and Custom Styles */
        body { font-family: 'Inter', sans-serif; }
        .canvas-card { background-color: #ffffff; border-radius: 0.75rem; box-shadow:
0 4px 6px -1px rgb(0 0 0 / 0.1), 0 2px 4px -2px rgb(0 0 0 / 0.1); padding: 1.5rem;
height: 100%; border-top: 4px solid #4f46e5; }

        /* Process Flow Diagram Styles */
        .process-flow { display: flex; align-items: stretch; gap: 0.5rem; overflow-x:
auto; padding-bottom: 1rem; margin-top: 1rem; }
        .process-step { flex: 0 0 160px; display: flex; flex-direction: column;
justify-content: center; align-items: center; padding: 0.75rem; border: 1px solid
#d1d5db; border-radius: 0.5rem; text-align: center; position: relative; font-size:
0.85rem; min-height: 100px; }
        .process-step.pain-point { border-color: #ef4444; background-color: #fee2e2; }
        .process-step.future-step { border-color: #22c55e; background-color: #dcfce7;
}
        .process-arrow { display: flex; justify-content: center; align-items: center;
color: #9ca3af; font-size: 1.5rem; font-weight: bold; }

        .tag { display: inline-block; padding: 0.25rem 0.5rem; border-radius: 9999px;
font-size: 0.75rem; font-weight: 600; margin-bottom: 0.5rem; }
        .tag-persona { background-color: #e0e7ff; color: #4338ca; }
        .tag-jtbd { background-color: #fef3c7; color: #92400e; }

        /* Print Styles */
        @media print {
            body { -webkit-print-color-adjust: exact !important; print-color-adjust:
exact !important; background-color: white !important; }
            #header { display: none !important; }
```

```
            .canvas-card { box-shadow: none; border: 1px solid #e5e7eb;
page-break-inside: avoid; margin-bottom: 1rem; }
            .process-flow { display: flex; flex-wrap: wrap; }
        }
    </style>
</head>
<body class="bg-gray-50 text-gray-800 p-4 sm:p-6 lg:p-8">
    <header id="header" class="flex justify-between items-center mb-6">
        <div>
           <h1 class="text-2xl font-bold text-gray-900">Garden App: Business Process
Canvas</h1>
           <p class="text-gray-500">Strategic Flow for Suburban Newbie Homeowners</p>
        </div>
        <button id="downloadButton" class="bg-indigo-600 text-white font-bold py-2
px-4 rounded-lg hover:bg-indigo-700 focus:outline-none focus:ring-2
focus:ring-offset-2 focus:ring-indigo-500">Download as PDF</button>
    </header>
    <main>
       <div class="grid grid-cols-1 md:grid-cols-2 gap-6 mb-8">
           <div class="canvas-card">
               <div class="flex justify-between items-start">
                   <span class="tag tag-persona">Persona: Maya & Liam</span>
                   <span class="text-xs font-bold text-indigo-600">FEATURE 1</span>
               </div>
               <h2 class="text-xl font-bold mb-2">Plant Scanner & Health Check</h2>
               <p class="text-sm text-gray-600 mb-4">AI-powered visual identification
and health diagnosis tool for immediate "Fear of Killing" mitigation.</p>
               <div class="mb-4">
                   <h3 class="text-xs font-bold text-gray-400 uppercase
tracking-widest mb-1">Benefit</h3>
                   <p class="text-sm font-medium">Instant identification eliminates
"What is this?" anxiety and identifies pests before damage spreads.</p>
               </div>
               <div class="process-flow">
                   <div class="process-step pain-point">Sees unknown plant or bug
(Pain)</div>
                   <div class="process-arrow">→</div>
                   <div class="process-step future-step"><strong>User:</strong> Opens
Scanner & Takes Photo</div>
                   <div class="process-arrow">→</div>
                   <div class="process-step"><strong>System:</strong> AI Analysis &
Species Match</div>
```

```
                    <div class="process-arrow">→</div>
                    <div class="process-step"><strong>System:</strong> Displays Health
Score & ID</div>
                </div>
            </div>

            <div class="canvas-card">
                <div class="flex justify-between items-start">
                    <span class="tag tag-persona">Persona: Maya</span>
                    <span class="text-xs font-bold text-indigo-600">FEATURE 2</span>
                </div>
                <h2 class="text-xl font-bold mb-2">Localized Maintenance Calendar</h2>
                <p class="text-sm text-gray-600 mb-4">Dynamic scheduling that adapts to
hyper-local weather sensors and specific plant needs.</p>
                <div class="mb-4">
                    <h3 class="text-xs font-bold text-gray-400 uppercase
tracking-widest mb-1">Benefit</h3>
                    <p class="text-sm font-medium">Removes the "Mental Load" of
remembering care tasks; prevents over/under watering.</p>
                </div>
                <div class="process-flow">
                    <div class="process-step"><strong>System:</strong> Detects Local
Rain Forecast</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step future-step"><strong>System:</strong>
Updates Daily Task List</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step"><strong>User:</strong> Receives "Skip
Watering" Alert</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step"><strong>User:</strong> Marks Task as
Complete</div>
                </div>
            </div>

            <div class="canvas-card">
                <div class="flex justify-between items-start">
                    <span class="tag tag-persona">Persona: Maya</span>
                    <span class="text-xs font-bold text-indigo-600">FEATURE 3</span>
                </div>
                <h2 class="text-xl font-bold mb-2">Nursery Onboarding (Inventory)</h2>
```

```
                <p class="text-sm text-gray-600 mb-4">Quick-add system for new
purchases to instantly integrate them into the home ecosystem.</p>
                <div class="mb-4">
                    <h3 class="text-xs font-bold text-gray-400 uppercase
tracking-widest mb-1">Benefit</h3>
                    <p class="text-sm font-medium">Protects the "sunk cost" of nursery
investments by providing immediate care instructions.</p>
                </div>
                <div class="process-flow">
                    <div class="process-step"><strong>User:</strong> Scans nursery tag
or plant</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step future-step"><strong>System:</strong>
Pulls Care Data (POWO API)</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step"><strong>User:</strong> Sets "Garden
Location"</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step"><strong>System:</strong> Syncs with
Maintenance Engine</div>
                </div>
            </div>

            <div class="canvas-card">
                <div class="flex justify-between items-start">
                    <span class="tag tag-persona">Persona: Liam</span>
                    <span class="text-xs font-bold text-indigo-600">FEATURE 4</span>
                </div>
                <h2 class="text-xl font-bold mb-2">Companion & Sunlight Guide</h2>
                <p class="text-sm text-gray-600 mb-4">Educational layer helping users
optimize plant placement for yield and pest control.</p>
                <div class="mb-4">
                    <h3 class="text-xs font-bold text-gray-400 uppercase
tracking-widest mb-1">Benefit</h3>
                    <p class="text-sm font-medium">Increases success rates for "Grow
Your Own" edible crops by grouping compatible species.</p>
                </div>
                <div class="process-flow">
                    <div class="process-step pain-point">Plants die from wrong light
(Pain)</div>
                    <div class="process-arrow">→</div>
```

```
                    <div class="process-step"><strong>User:</strong> Selects "Plot
Planning"</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step future-step"><strong>System:</strong>
Suggests Companion Plants</div>
                    <div class="process-arrow">→</div>
                    <div class="process-step"><strong>User:</strong> Drags plants to
map</div>
                </div>
            </div>
        </div>

        <div class="canvas-card">
            <h2 class="text-xl font-bold mb-6 flex items-center">
                <svg class="w-6 h-6 mr-2 text-indigo-600" fill="none"
stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M9 12l2 2 4-4m6 2a9 9 0 11-18 0 9 9 0 0118
0z"></path></svg>
                The Happy Path: End-to-End User Journey
            </h2>
            <div class="space-y-4">
                <div class="flex items-start">
                    <div class="bg-indigo-100 text-indigo-700 rounded-full h-8 w-8 flex
items-center justify-center font-bold mr-4 shrink-0">1</div>
                    <div>
                        <p class="font-bold">Initial Trigger & Setup</p>
                        <p class="text-sm text-gray-600"><strong>User Action:</strong>
Downloads app and enters zip code. <br><strong>System (App):</strong> Fetches local
frost dates and historical weather data via OpenWeatherMap API.</p>
                    </div>
                </div>
                <div class="flex items-start">
                    <div class="bg-indigo-100 text-indigo-700 rounded-full h-8 w-8 flex
items-center justify-center font-bold mr-4 shrink-0">2</div>
                    <div>
                        <p class="font-bold">Inventory Creation (The Scanner)</p>
                        <p class="text-sm text-gray-600"><strong>User Action:</strong>
Scans 5 plants in their backyard. <br><strong>System (AI Engine):</strong> Identifies
species and classifies them into "Sun-loving" vs "Shade-loving" categories.</p>
                    </div>
                </div>
                <div class="flex items-start">
```

```
                    <div class="bg-indigo-100 text-indigo-700 rounded-full h-8 w-8 flex
items-center justify-center font-bold mr-4 shrink-0">3</div>
                    <div>
                        <p class="font-bold">Automated Care Sync</p>
                        <p class="text-sm text-gray-600"><strong>System (App):</strong>
Populates a 7-day calendar. Alerts user: "Heavy rain tonight. No watering needed for
48 hours."</p>
                    </div>
                </div>
                <div class="flex items-start">
                    <div class="bg-indigo-100 text-indigo-700 rounded-full h-8 w-8 flex
items-center justify-center font-bold mr-4 shrink-0">4</div>
                    <div>
                        <p class="font-bold">Maintenance & Learning</p>
                        <p class="text-sm text-gray-600"><strong>User Action:</strong>
Receives "Deadheading" task. Clicks "Why?" <br><strong>System (App):</strong> Displays
15-second video explaining how it promotes more blooms.</p>
                    </div>
                </div>
            </div>

            <div class="mt-8 p-4 bg-gray-50 rounded-lg border border-dashed
border-gray-300">
                <h3 class="text-sm font-bold text-gray-800 mb-2">Edge Case Handling:
Identification Failure</h3>
                <p class="text-xs text-gray-600 italic">If the AI fails to identify a
plant (>85% confidence threshold), the <strong>System</strong> offers an "Expert Human
Review." The <strong>Internal Admin Staff</strong> receives the photo, IDs it manually
within 24 hours, and the <strong>System</strong> notifies the user once the ID is
confirmed.</p>
            </div>
        </div>
    </main>
    <script>
        document.addEventListener('DOMContentLoaded', function () {
            const downloadButton = document.getElementById('downloadButton');
            if(downloadButton) {
                downloadButton.addEventListener('click', () => window.print());
            }
        });
    </script>
</body>
```

```html
</html>
```